

B.Sc. (Hons.) SEMESTER III EXAMINATION 2016-17
STATISTICS

Paper No. STB301: Statistical Inference

Time: Three hours

Max. Marks: 70

Note: Attempt any five questions. All questions carry equal marks.

1. (a) Explain the concept of sampling distribution with an example. Suppose $X_1, X_2, \dots, X_n$ be a random sample of size n from a population with finite mean μ and variance σ^2 , then show that:

$$E(\bar{X}) = \mu \quad \text{and} \quad V(\bar{X}) = \frac{\sigma^2}{n}$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean.

- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from a Bernoulli distribution with parameter p , then show that the sampling distribution of $S = \sum_{i=1}^n X_i$ is a binomial distribution.
2. (a) Define probabilities of Type-I and Type-II errors. A sample of size 1 is taken from an exponential distribution with parameter λ . To test $H_0 : \lambda = 1$ against $H_1 : \lambda > 1$, suppose the critical region is considered as $x > 2$. Find the probabilities of Type-I and Type-II errors.
- (b) Define the null and alternative hypotheses with examples. Explain with illustrations how to specify the null and alternative hypotheses in a hypothesis testing problem.
3. (a) Write the density function of t -distribution. Mention its distributional properties. Also, define Student's t -statistic and its uses.
- (b) What is F -distribution? Explain the testing procedure for testing the equality of two variances of two independent normal populations.
4. (a) State the weak law of large numbers (WLLN). Let $X_1, X_2, \dots, X_n, \dots$ be a sequence of independent and identically distributed random variables with a Bernoulli distribution with parameter p , then show that the sequence obeys the WLLN.
- (b) State the central limit theorem. Suppose $X_1, X_2, \dots, X_n$ be a random sample of size n from a population with finite mean 5 and variance 16.5. When n is large, say 50, obtain the probability $P[\bar{X} < 5]$.
5. (a) Define order statistics with examples in real-life applications. Show that the r^{th} order statistic from a uniform distribution $U[0, 1]$ follows a beta distribution.
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and variance σ^2 . Show that the sampling distribution of the mean is also a normal distribution.
6. (a) What are the advantages and disadvantages of non-parametric tests over parametric tests?
- (b) Write the Mann-Whitney-Wilcoxon test statistic. To compare the variability of two brands of tires, the following mileages (in 1000 miles) were obtained for eight tires of each kind:

Brand A: 32.1, 20.6, 17.8, 28.4, 19.6, 21.4, 19.9, 30.1

Brand B: 19.8, 27.6, 30.8, 27.6, 34.1, 18.7, 16.9, 17.9

Test the null hypothesis that the two samples come from the same population using the Mann-Whitney-Wilcoxon test.

7. Write short notes on any **three** of the following:

- (a) Uses of chi-square distribution
 - (b) Median test
 - (c) p-value
 - (d) Test of independence of two attributes
 - (e) Fisher's Z transformation
8. (a) Explain the procedure of testing for the significance of the sample correlation coefficient in random sampling from a bivariate normal distribution.
- (b) Nine students held their breath, once after breathing normally and relaxing for one minute, and once after hyperventilating for one minute. The table indicates how long (in sec) they were able to hold their breath:

Subject	A	B	C	D	E	F	G	H	I
Normal	56	56	65	65	50	25	87	44	35
Hypervent	87	91	85	91	75	28	122	66	58

Mentioning the null and alternative hypotheses clearly, test the significance of the correlation coefficient given $t_{9,0.05} = 2.262$ and $t_{7,0.05} = 2.365$.

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>